

Derivatives

$$\begin{aligned}\frac{d}{dx}(\sin^{-1} x) &= \frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx}(\cos^{-1} x) &= -\frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx}(\tan^{-1} x) &= \frac{1}{1+x^2} \\ \frac{d}{dx}(\sinh^{-1} x) &= \frac{1}{\sqrt{1+x^2}} \\ \frac{d}{dx}(\cosh^{-1} x) &= \frac{1}{\sqrt{x^2-1}} \\ \frac{d}{dx}(\tanh^{-1} x) &= \frac{1}{1-x^2}\end{aligned}$$

Integrals

$$\begin{aligned}\int u dv &= uv - \int v du \\ \int \tan x dx &= \ln |\sec x| + C \\ \int \sec x dx &= \ln |\sec x + \tan x| + C \\ \int \frac{dx}{x^2+a^2} &= \frac{1}{a} \tan^{-1}(x/a) + C\end{aligned}$$

Approximate Integration

$$\begin{aligned}|E_T| &\leq \frac{K(b-a)^3}{12n^2} \\ |E_M| &\leq \frac{K(b-a)^3}{24n^2} \\ |E_S| &\leq \frac{L(b-a)^5}{180n^4}\end{aligned}$$

Hyperbolic Functions

$$\begin{aligned}\cosh x &= \frac{e^x + e^{-x}}{2} \\ \sinh x &= \frac{e^x - e^{-x}}{2} \\ \tanh x &= \frac{\sinh x}{\cosh x}\end{aligned}$$

$$\begin{aligned}\sinh^{-1} x &= \ln(x + \sqrt{x^2 + 1}) \\ \cosh^{-1} x &= \ln(x + \sqrt{x^2 - 1}) \\ \tanh^{-1} x &= \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)\end{aligned}$$

Trig & Hyperbolic Identities

$$\begin{aligned}\cos^2 x + \sin^2 x &= 1 \\ \tan^2 x + 1 &= \sec^2 x \\ \sin^2 x &= \frac{1}{2}(1 - \cos 2x)\end{aligned}$$

Area Between Curves

$$A = \int_a^b (f - g) dx$$

Volumes

$$\begin{aligned}V &= \int A(x) dx \\ V &= \int 2\pi x f(x) dx\end{aligned}$$

Arc Length

$$L = \int_a^b \sqrt{1 + f'^2} dx$$

Surface Area

$$\begin{aligned}A &= \int 2\pi f \sqrt{1 + f'^2} dx \\ A &= \int 2\pi x \sqrt{1 + f'^2} dx\end{aligned}$$

Moments and Centroid

$$\begin{aligned}M_x &= \frac{\rho}{2} \int_a^b (f^2 - g^2) dx \\ M_y &= \rho \int_a^b x(f - g) dx \\ m &= \rho \int_a^b (f - g) dx \\ (\bar{x}, \bar{y}) &= (M_y/m, M_x/m)\end{aligned}$$

Work

$$W = \int_a^b F(x) dx$$

Hydrostatic Force

$$F = \rho g \int w(y) d(y) dy$$

Polar Coordinates

$$\begin{aligned}\frac{dy}{dx} &= \frac{r' \sin \theta + r \cos \theta}{r' \cos \theta - r \sin \theta} \\ A &= \frac{1}{2} \int_a^b (f^2 - g^2) d\theta \\ L &= \int_a^b \sqrt{r^2 + (r')^2} d\theta\end{aligned}$$

Series

$$\begin{aligned}\sum_{n=0}^{\infty} ar^n &= \frac{a}{1-r}, |r| < 1 \\ |s_n - S| &\leq b_{n+1} \\ c_n &= \frac{f^{(n)}(a)}{n!} \\ R_n(x) &= \frac{f^{(n+1)}(z)}{(n+1)!} (x-a)^{n+1} \\ \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n \\ e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \\ \sin x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \\ \cos x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \\ \tan^{-1} x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \\ \ln(1+x) &= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} \\ (1+x)^k &= \sum_{n=0}^{\infty} \binom{k}{n} x^n\end{aligned}$$

